Fields and Waves I

Lecture 13

Laplace's Equation, Poisson's Equation and Finite Difference Method

Alex Patterson

Electrical, Computer, and Systems Engineering Department

Rensselaer Polytechnic Institute, Troy, NY

These slides were prepared through the work of the following people:

- Kenneth A. Connor – ECSE Department, Rensselaer Polytechnic Institute, Troy, NY (Emeritus)
- J. Darryl Michael – GE Global Research Center, Niskayuna, NY
- Thomas P. Crowley – National Institute of Standards and Technology, Boulder, CO
- Sheppard J. Salon – ECSE Department, Rensselaer Polytechnic Institute, Troy, NY (Emeritus)
- Lale Ergene – ITU Informatics Institute, Istanbul, Turkey
- Jeffrey Braunstein – ECE Department, University at Albany
- James Lu - ECSE Department, Rensselaer Polytechnic Institute, Troy, NY
- James Dylan Rees - ECSE Department, Rensselaer Polytechnic Institute, Troy, NY
- Alex Patterson - ECSE Department, Rensselaer Polytechnic Institute, Troy, NY

Materials from other sources are referenced where they are used.
Those listed as Ulaby are figures from Ulaby's textbook.

Overview

- Laplace’s and Poisson’s Equations
- Finite Difference Method
- Finite Element Method

Laplace’s and Poisson’s Equations

Laplace and Poisson Equations

- Integral Form

$$\oint \vec{D} \cdot \vec{ds} = \int \rho \cdot dv$$

$$\oint \vec{E} \cdot \vec{dl} = 0$$

- Differential Form

$$\nabla \cdot \vec{D} = \rho$$

$$\nabla \times \vec{E} = 0$$

What if we try to rewrite Maxwell's Equations in terms of voltage?

$$\vec{E} = -\nabla V$$

Laplace and Poisson Equations

- First, the curl equation

$$\vec{E} = -\nabla V \quad \Rightarrow \quad \nabla \times \vec{E} = 0$$

since $\nabla \times (\nabla f) = 0$

Laplace and Poisson Equations

- Next, the divergence equation

$$\nabla \cdot \vec{D} = \nabla \cdot (\epsilon \vec{E}) = -\epsilon(\nabla \cdot (\nabla V)) = \rho$$

$$\Rightarrow \nabla^2 V = -\frac{\rho}{\epsilon}$$

Laplace and Poisson Equations

- Poisson's Equation:

$$\nabla^2 V = -\frac{\rho}{\epsilon}$$

- Laplace's Equation:
  (Special case of Poisson's equation)

$$\nabla^2 V = 0$$

Laplace and Poisson Equations

- ∇^2 is called the **Laplacian** operator. In Cartesian coordinates:

$$\nabla^2 V = \left[\frac{\partial V}{\partial x} \quad \frac{\partial V}{\partial y} \quad \frac{\partial V}{\partial z}\right] \bullet \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

Laplace and Poisson Equations

- Cylindrical Laplacian Operator:

$$\nabla^2 V = \frac{1}{r}\frac{\partial}{\partial r}(r\frac{\partial V}{\partial r}) + \frac{1}{r^2}\frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

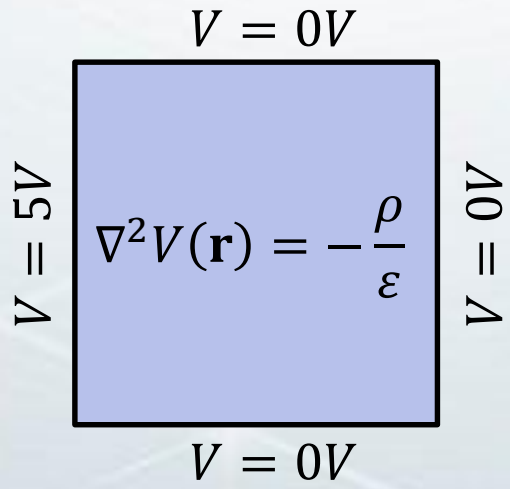
- Spherical Laplacian Operator:

$$\nabla^2 V = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial V}{\partial r}) + \frac{1}{r^2 sin\theta}\frac{\partial}{\partial \theta}(sin\theta\frac{\partial V}{\partial \theta}) + \frac{1}{r^2 sin^2\theta}\frac{\partial^2 V}{\partial \phi^2}$$

Laplace and Poisson Equations

- Laplace's equation and Poisson's equation are general mathematical expressions that allow us to solve scalar fields if we know the **boundary conditions**.

- They are used for solving for:
  - Voltage
  - Heat
  - Gravity
  - Aspects of fluid flow
  - Various abstract mathematical fields
  - etc.



Laplace and Poisson Equations

- In General

$$D_{1n} - D_{2n} = \rho_s \qquad \vec{E}_{1t} = \vec{E}_{2t}$$

- Dielectric-Dielectric

$$D_{1n} = D_{2n} \qquad \vec{E}_{1t} = \vec{E}_{2t}$$

- Conductor-Dielectric

$$D_{1n} = \rho_s \qquad \vec{E}_{1t} = 0$$

Laplace and Poisson Equations

- Dielectric-Dielectric

	Normal To Boundary	Tangential To Boundary
Fields	$D_{1n} = D_{2n}$	$E_{1t} = E_{2t}$
Potential	$\epsilon_1 \frac{\partial V_1}{\partial n} = \epsilon_2 \frac{\partial V_2}{\partial n}$	$V_1 = V_2$

$$\vec{D} = \varepsilon\vec{E} = -\varepsilon\nabla V$$

Laplace and Poisson Equations

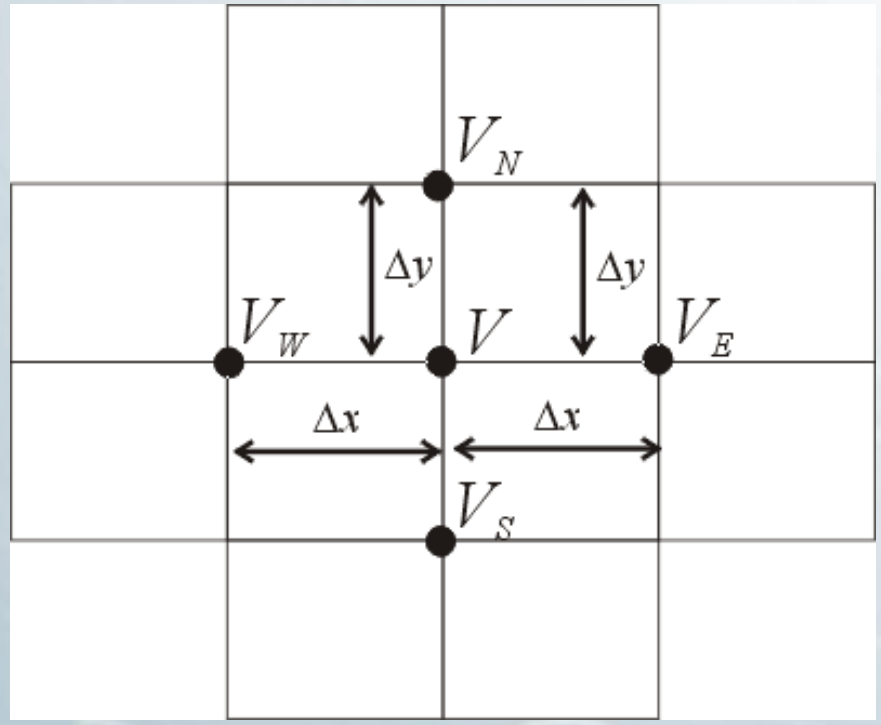
- Dielectric-Conductor (assuming conductor is material 2)

	Normal To Boundary	Tangential To Boundary
Fields	$D_{1n} = \rho_s$	$E_{1t} = 0$
Potential	$\epsilon_1 \frac{\partial V_1}{\partial n} = \rho_s$	$V_1 = constant$

Expression would have a "-" sign
if conductor were material 1

Laplace and Poisson Equations

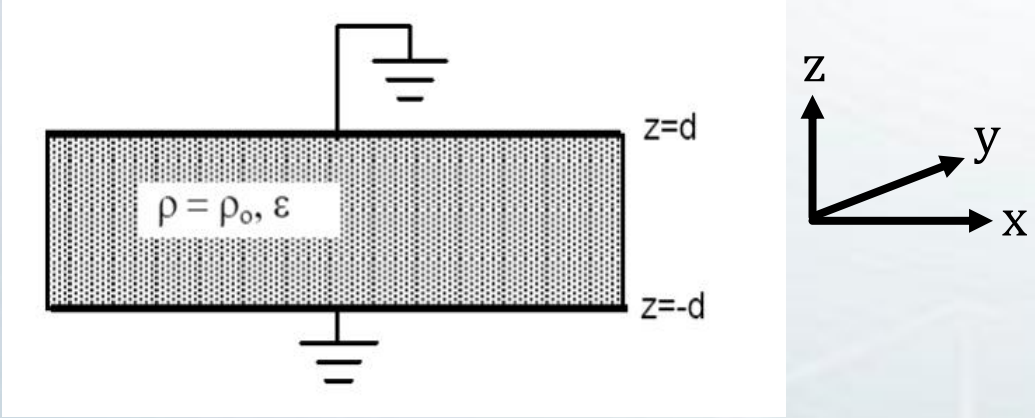
- This is the most common way of finding electric fields, since it is easier to implement numerically on a grid



https://propagation.ece.gatech.edu/ECE3025/tutorials/ComputingLaplace/DiscreteLaplace.htm

Laplace and Poisson Equations

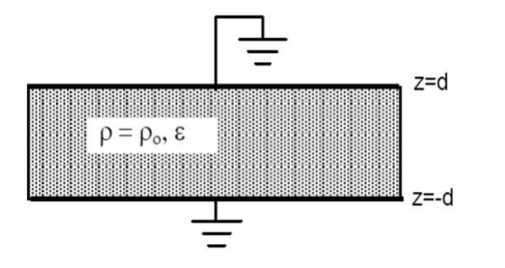
A charged region of a semiconductor is sandwiched between two grounded conductors as shown below.



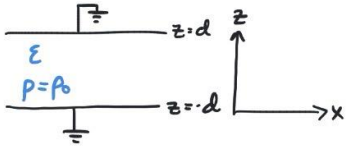
1. Solve for V(z) directly using Poisson's Equation
2. Find **E** and **D**
3. Find the charge density on the conductors

Laplace and Poisson Equations

1. Solve for V(z) directly using Poisson's Equation



Solving for V(z) using Poisson's equation



Solve for V(z): $\nabla^2 V(\vec{r}) = -\frac{\rho_0}{\varepsilon}$ ← cartesian coordinates

$\nabla^2 V(\vec{r}) = \frac{\partial^2 V}{\partial x^2}$ ($\partial E_x = 0$) $+ \frac{\partial^2 V}{\partial y^2}$ ($\partial E_y = 0$) $+ \frac{\partial^2 V}{\partial z^2} = \frac{\partial^2 V(z)}{\partial z^2}$

$\rightarrow \frac{\partial^2 V(z)}{\partial z^2} = -\frac{\rho_0}{\varepsilon}$

1) Solve for the general solution:

$$\int \frac{\partial^2 V(z)}{\partial z} dz = -\int \frac{\rho_0}{\varepsilon} dz \rightarrow \frac{\partial V(z)}{\partial z} = -\frac{\rho_0}{\varepsilon} z + C_0$$

$$\int \frac{\partial V(z)}{\partial z} dz = \int \left(-\frac{\rho_0}{\varepsilon} z + C_0\right) dz \rightarrow V(z) = -\frac{\rho_0}{2\varepsilon} z^2 + C_0 z + C_1$$

2) Find the particular solution by applying boundary conditions

#1: $V(z=d) = 0V$
#2: $V(z=-d) = 0V$

$$\rightarrow \begin{cases} (1)\ 0 = -\frac{\rho_0}{2\varepsilon} d^2 + C_0 d + C_1 \\ (2)\ 0 = -\frac{\rho_0}{2\varepsilon} d^2 - C_0 d + C_1 \end{cases}$$

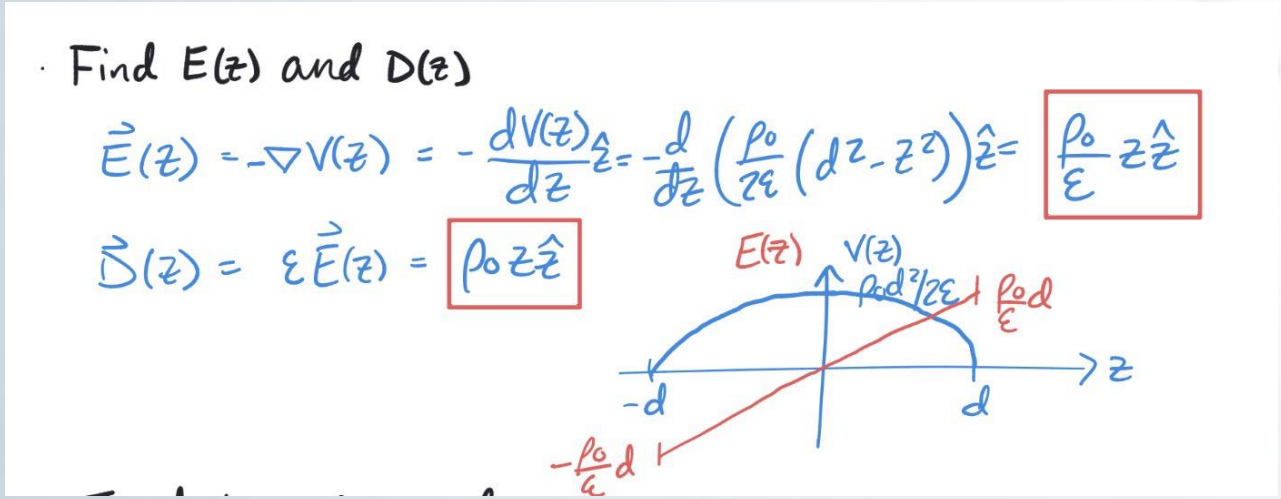
Solve for C_1: $(1)+(2) \rightarrow 0 = -\frac{\rho_0}{\varepsilon} d^2 + 2C_1 \rightarrow C_1 = \frac{\rho_0}{2\varepsilon} d^2$

$(1)-(2) \rightarrow 0 = 2C_0 d \rightarrow C_0 = 0$

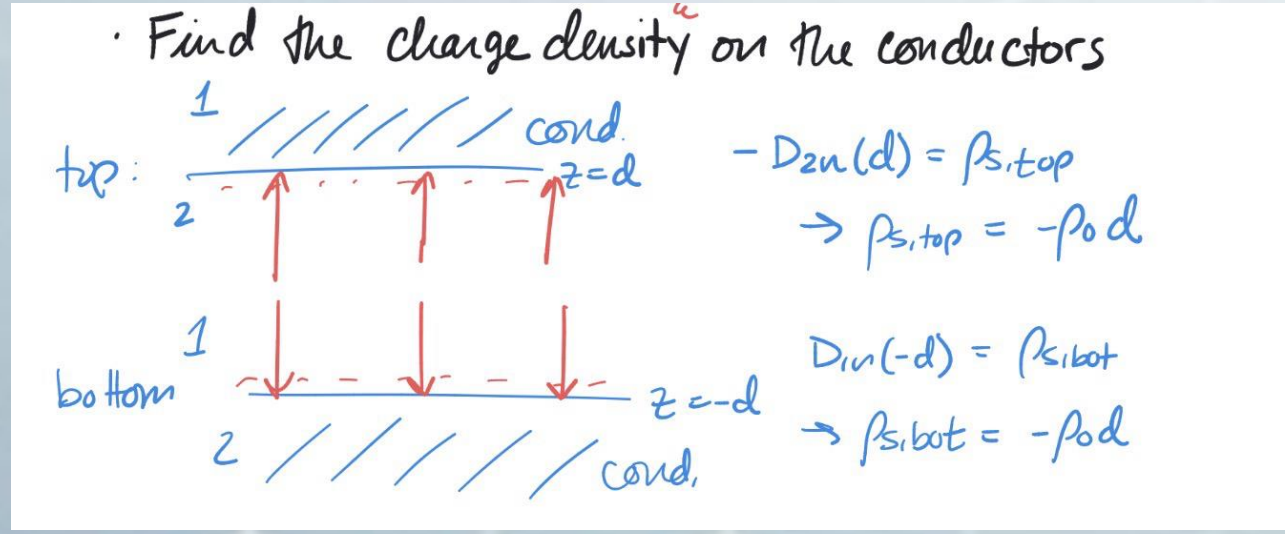
$$V(z) = -\frac{\rho_0}{2\varepsilon} z^2 + \frac{\rho_0}{2\varepsilon} d^2 = \boxed{\frac{\rho_0}{2\varepsilon}(d^2 - z^2)}$$

Laplace and Poisson Equations

2. Find **E** and **D**



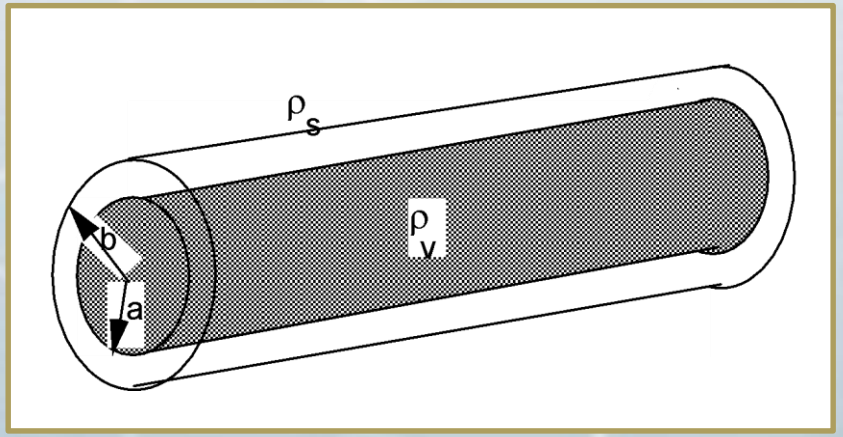
3. Find the charge density on the conductors



Laplace and Poisson Equations

A coaxial cable has an inner conductor (at r = *a*) held at voltage V_0 and an outer conductor (at r = *b*) that is grounded. There is no charge other than the surface charge on the conductors.

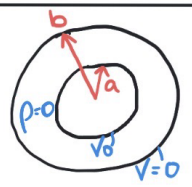
1. Solve for V(r) directly using Laplace's Equation
2. Solve for **E** and **D**
3. What is the charge density on the two conductors?
4. What is the capacitance per unit length?



Laplace and Poisson Equations

1. Solve for V(r) directly using Laplace's Equation

Potential in a coaxial cable



cylindrical coordinates

$V(r=a) = V_0$

$V(r=b) = 0$

$\rho = 0$ for $a<r<b$

- Solve for V(r) using Laplace's equation

$\nabla^2 V(\vec{r}) = 0 \rightarrow \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) = 0$

a) find the general solution
b) find the particular solution

a) $\frac{1}{r}\frac{d}{dr}\left(r\frac{dV}{dr}\right) = 0 \rightarrow \int \frac{d}{dr}\left(r\frac{dV}{dr}\right)dr = \int 0\,dr \rightarrow r\frac{dV}{dr} = C_0 \rightarrow \int \frac{dV}{dr}dr = C_0\int 1/r\,dr$

$\boxed{V(z) = C_0 \ln(r) + C_1}$

b) $\begin{cases} (1)\ V_0 = C_0\ln(a) + C_1 \\ (2)\ 0 = C_0\ln(b) + C_1 \end{cases} \rightarrow$

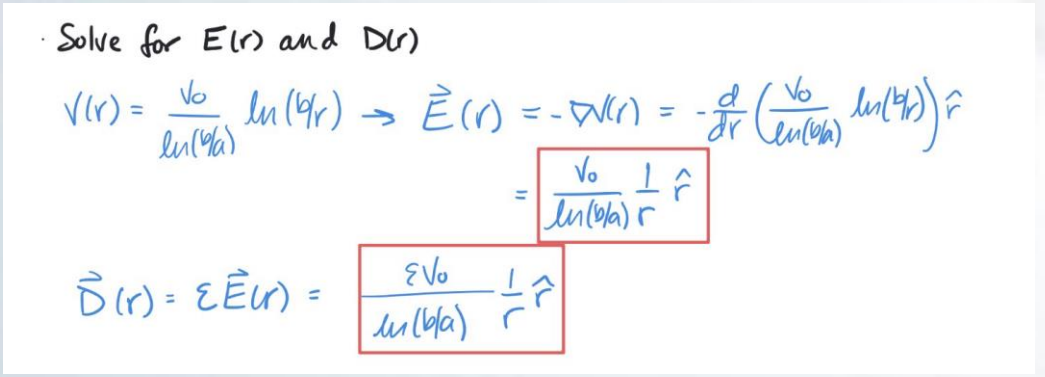
$(1)-(2): V_0 = C_0\ln(a/b) \rightarrow C_0 = \frac{V_0}{\ln(a/b)}$

$C_0 \rightarrow (2): 0 = \frac{V_0}{\ln(a/b)}\ln(b) + C_1 \rightarrow C_1 = -\frac{V_0}{\ln(a/b)}\ln(b)$

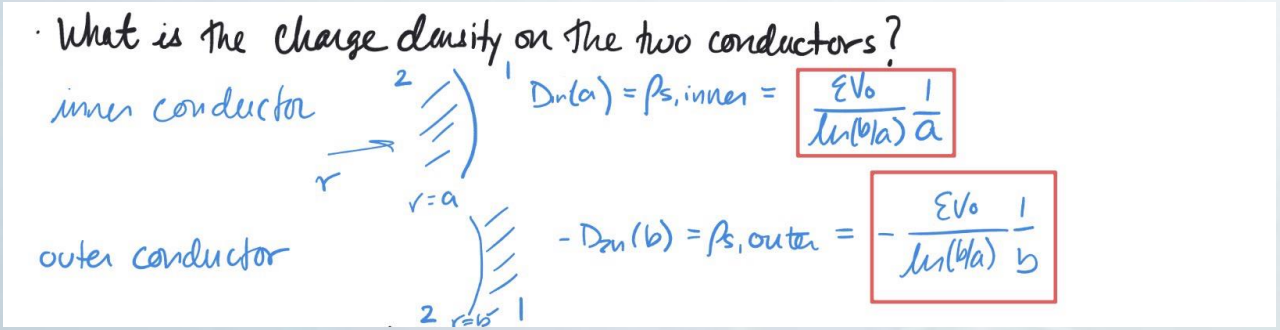
$V(z) = \frac{V_0}{\ln(a/b)}\ln(r) - \frac{V_0}{\ln(a/b)}\ln(b) = \frac{V_0}{\ln(a/b)}\ln(r/b) = \boxed{\frac{V_0}{\ln(b/a)}\ln(b/r)}$

Laplace and Poisson Equations

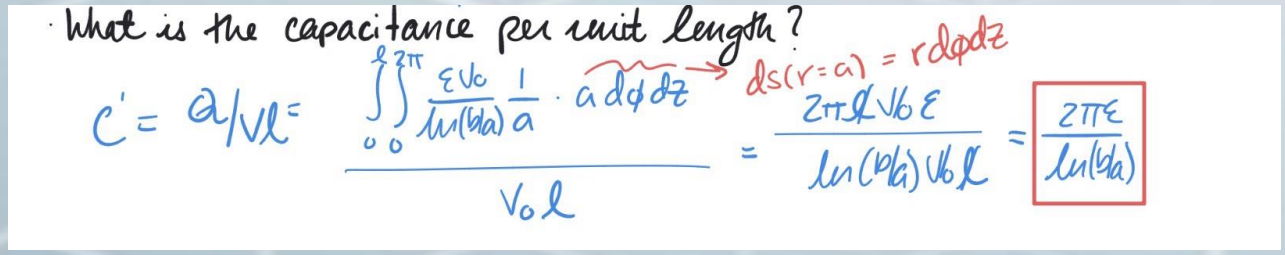
2. Solve for **E** and **D**



3. What is the charge density on the two conductors?



4. What is the capacitance per unit length?



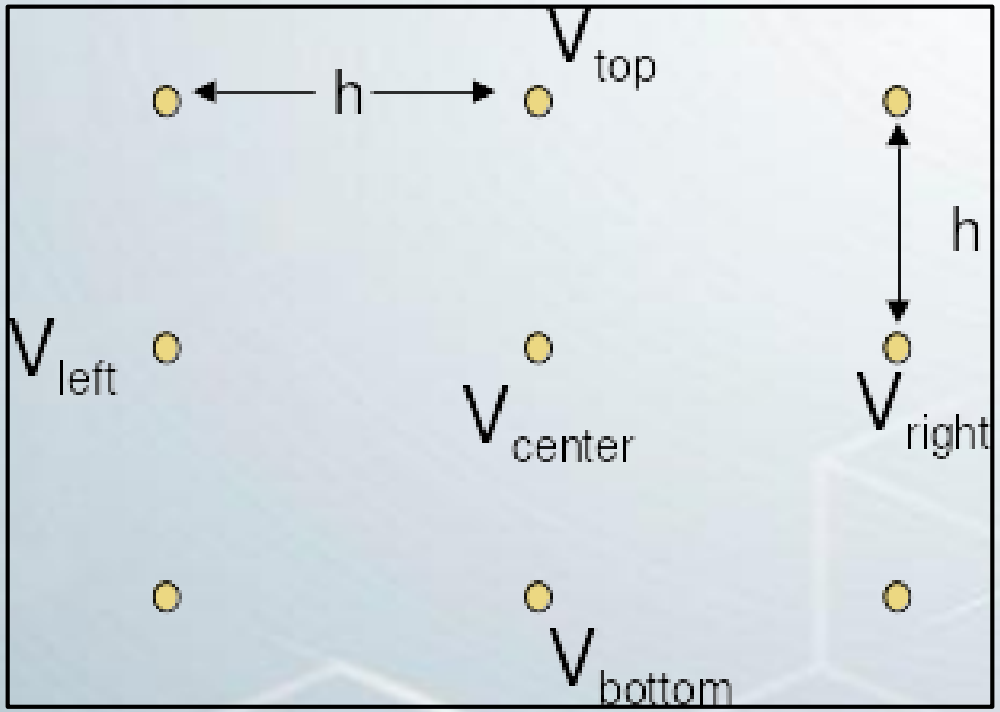
Laplace and Poisson Equations

- ...but as engineers, you all know that integrals can be very difficult to evaluate for all but very simple geometries.
- So how do we solve for V(r) when the geometry is more complex?
- We rely on numerical methods
  - Finite Difference
  - Finite Elements
  - Method of Moments
  - Etc.

Finite Difference Method for Solving Poisson's Equation

Finite Difference

Consider a grid of points in a voltage field with distance h between them.

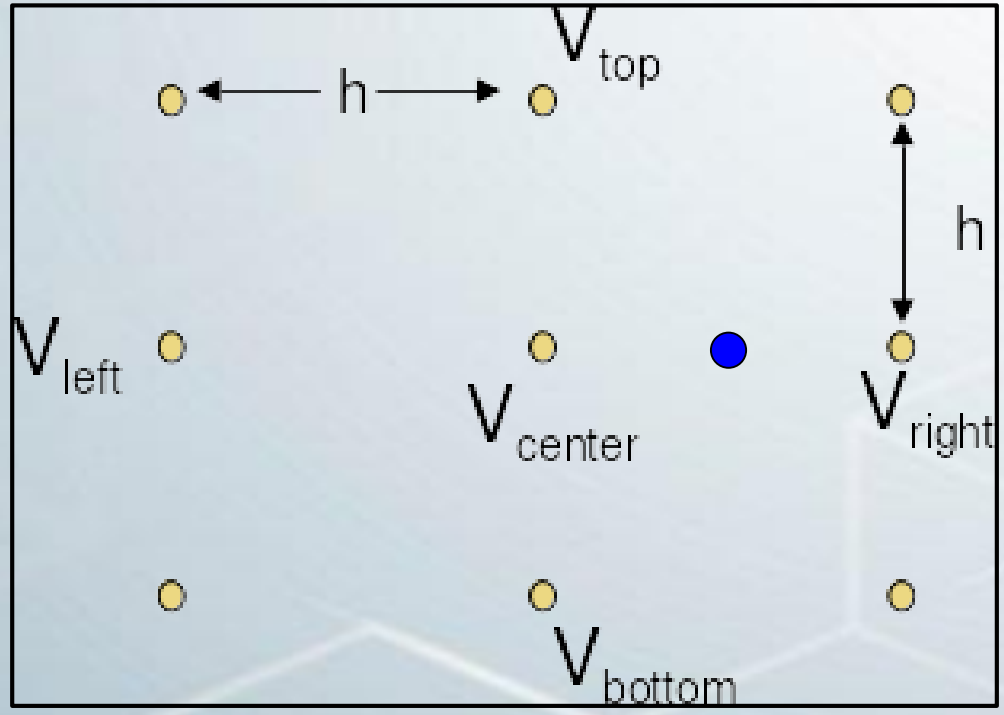


V_{center} is at the origin (0,0).

Finite Difference

At $(x, y) = (h/2, 0)$

$$E_x = -\frac{\partial V}{\partial x} \approx -\frac{\Delta V}{\Delta x} = -\frac{(V_{right} - V_{center})}{h}$$

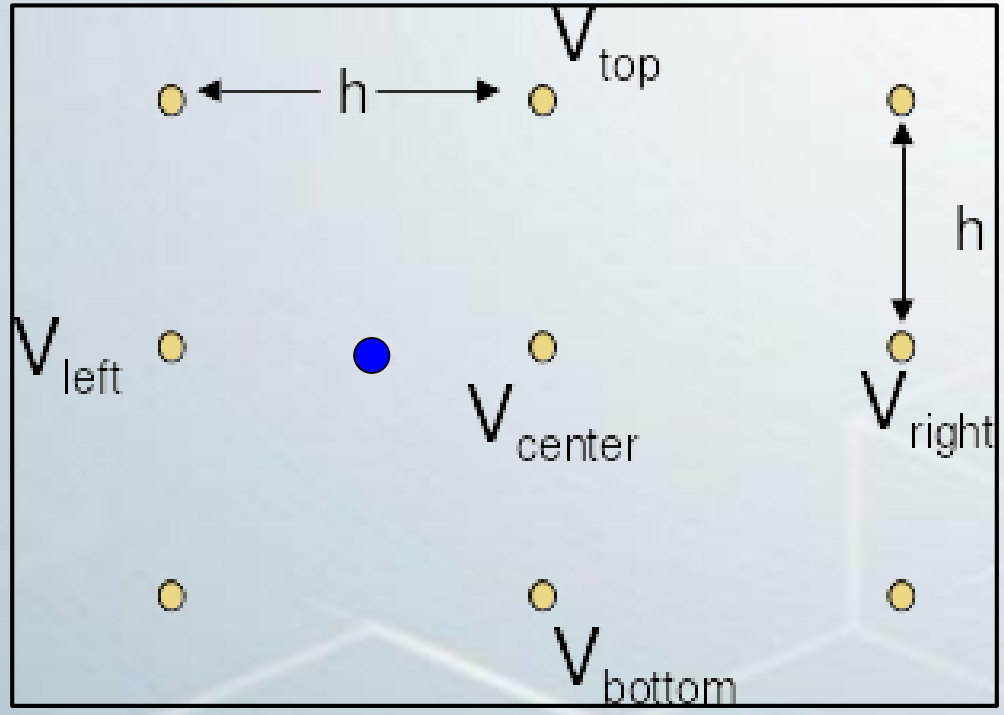


The E-field can be approximated by taking the difference of the two adjacent voltages.

Finite Difference

At $(x, y) = (-h/2, 0)$

$$E_x = -\frac{\partial V}{\partial x} \approx -\frac{\Delta V}{\Delta x} = -\frac{(V_{center} - V_{left})}{h}$$



The E-field can be approximated by taking the difference of the two adjacent voltages.

Finite Difference

- The derivative of the E-field can be approximated by taking the difference of the two adjacent E-field values

$$\nabla^2 V = \nabla \cdot (\nabla \mathrm{V}) = \nabla \cdot \vec{E} = -\frac{\partial E_x}{\partial x} - \frac{\partial E_y}{\partial y} - \frac{\partial E_z}{\partial z}$$

- To find the x-direction contribution to the E-field at V_{center}:

$$\frac{\partial E_x}{\partial x} \approx \frac{\Delta E_x}{\Delta x} = \frac{E_x\left(\frac{h}{2},0\right) - E_x\left(-\frac{h}{2},0\right)}{h} = \frac{2 \cdot V_{center} - V_{right} - V_{left}}{h^2}$$

- We can find a similar expression for the y-direction contribution:

$$\frac{\partial \vec{E}_y}{\partial y} \approx \frac{\Delta \vec{E}_y}{\Delta y} = \frac{2 \cdot V_{center} - V_{top} - V_{bottom}}{h^2}$$

Finite Difference

- Finally we obtain the following expression:

$$\nabla^2 V = \frac{4 \cdot V_{center} - (V_{right} + V_{left} + V_{top} + V_{bottom})}{h^2} = -\frac{\rho}{\varepsilon}$$

- Rearrange the equation to solve for V_{center}:

$$V_{center} = \frac{1}{4} \cdot \left(\sum V_{neighbors} - \frac{\rho \cdot h^2}{\varepsilon} \right)$$



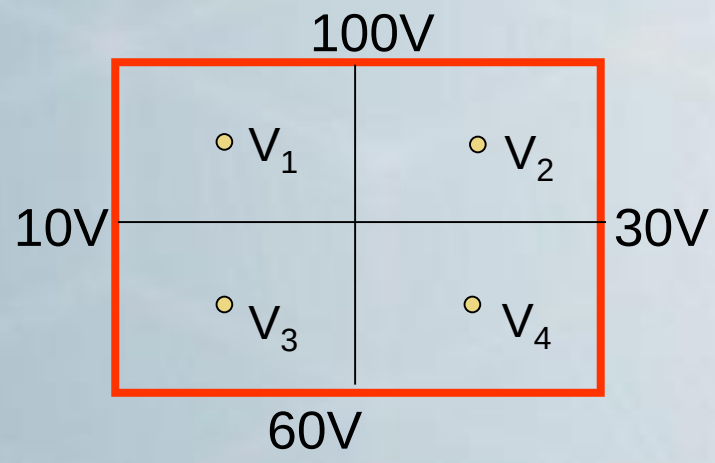
Poisson Equation Solver

$$V_{center} = \frac{1}{4} \cdot \left(\sum V_{neighbors} \right)$$



Laplace Equation Solver

Finite Difference



Solution Technique - by Iteration

Guess an initial solution: V=0 everywhere except boundaries

$$V_1 = V_2 = V_3 = V_4 = 0$$

Start:

Put new values back

$$V_1 = \tfrac{1}{4}\cdot(100+10+V_2+V_3) \longrightarrow V_1 = \tfrac{1}{4}\cdot(110+0) = 27.5$$

$$V_2 = \tfrac{1}{4}\cdot(100+30+V_1+V_4) \longrightarrow V_2 = \tfrac{1}{4}\cdot(130+27.5+0) = 39.375$$

$$V_3 = \tfrac{1}{4}\cdot(10+60+V_1+V_4) \longrightarrow V_3 = \tfrac{1}{4}\cdot(70+27.5+0) = 24.375$$

$$V_4 = \tfrac{1}{4}\cdot(30+60+V_2+V_3) \longrightarrow V_4 = \tfrac{1}{4}\cdot(90+39.375+24.375) = 38.4375$$

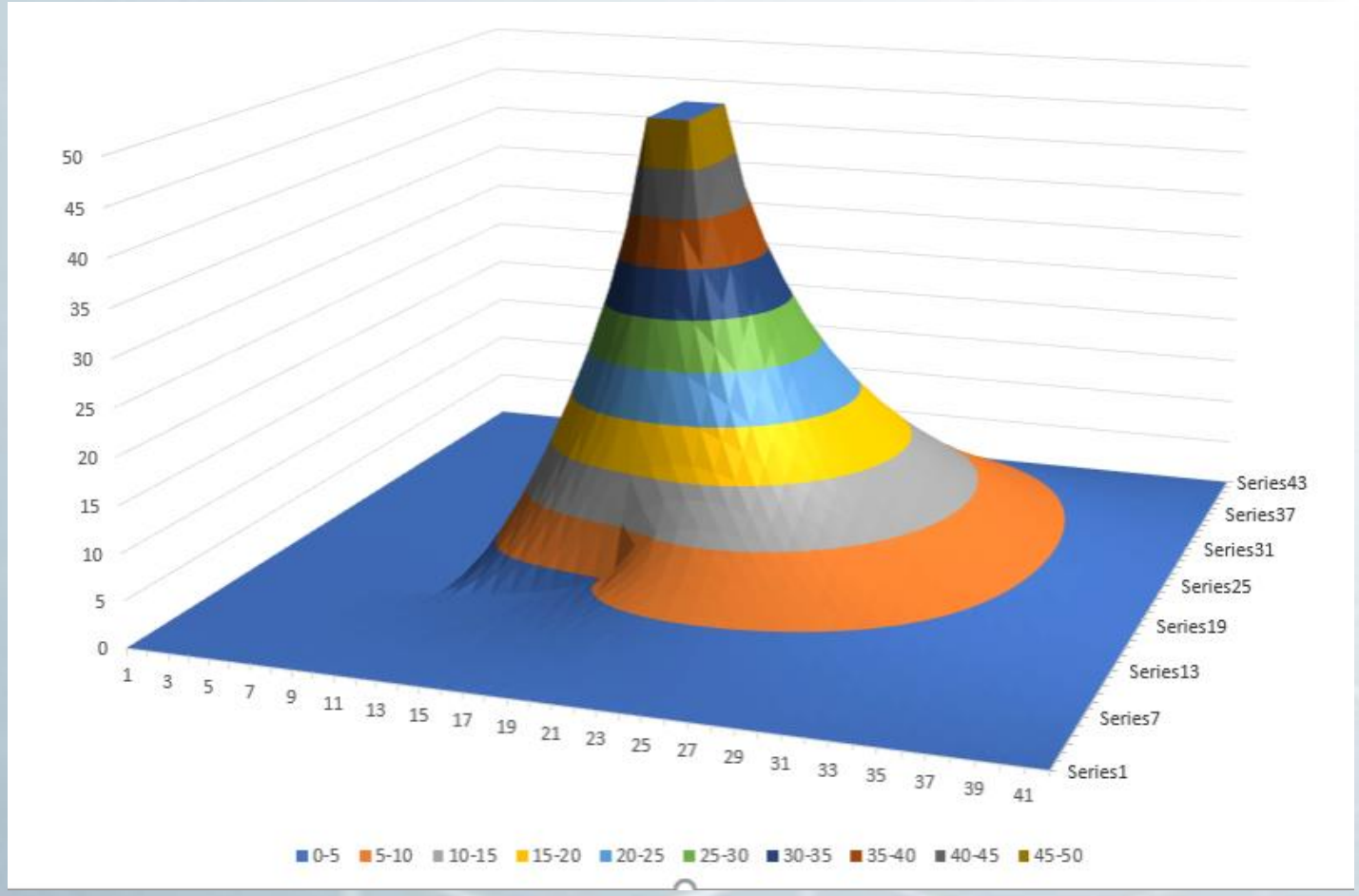
Do Lecture 13 Exercise in groups of up to 4.

Finite Difference

- The Finite Difference method can be done in a spreadsheet (one of our Studio Sessions will be based on this)



Finite Difference

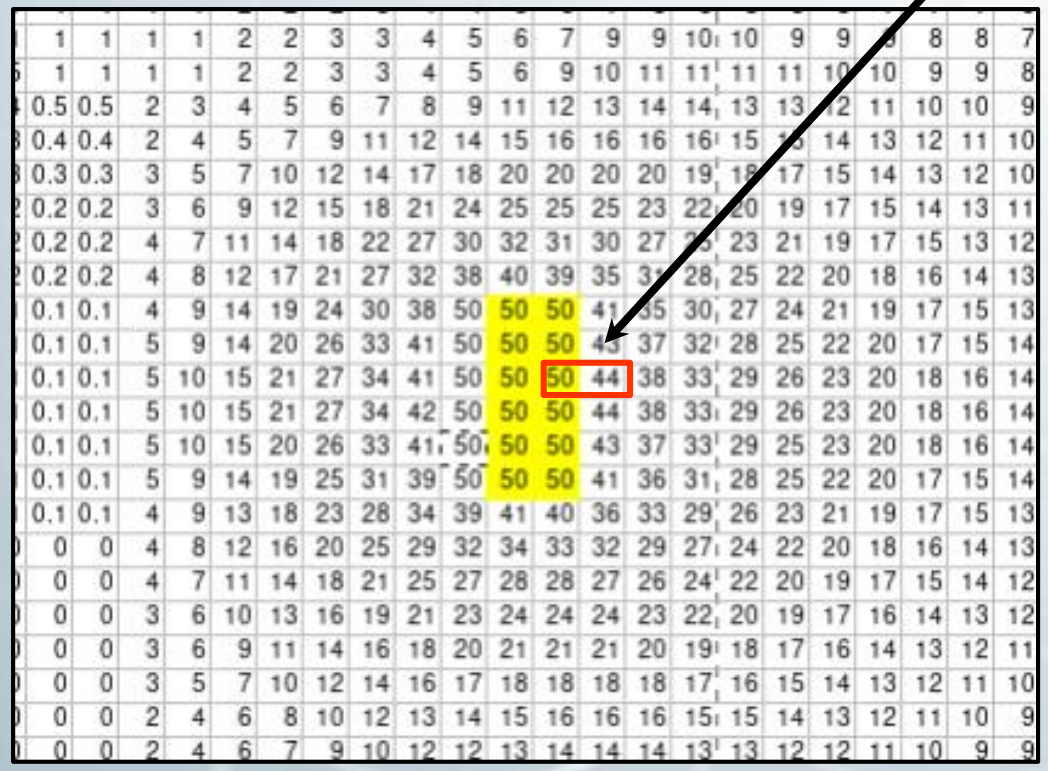


Surface chart made using Microsoft Excel

Finite Difference

- Suppose that this represents a conductive rectangle held at 50V.

$$E_x = -\frac{\partial V}{\partial x} \approx -\frac{\Delta V}{\Delta x}$$

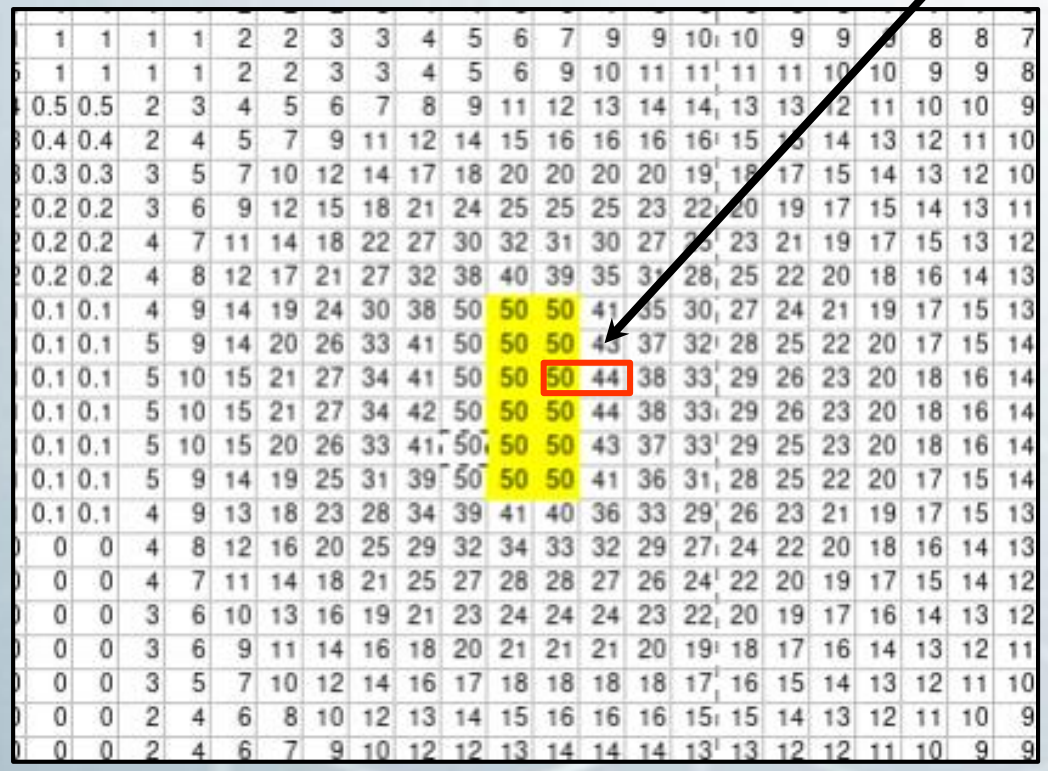


	1	1	1	1	2	2	3	3	4	5	6	7	9	9	10	10	9	9	8	8	7
	1	1	1	1	2	2	3	3	4	5	6	9	10	11	11	11	11	10	10	9	9	8
	0.5	0.5	2	3	4	5	6	7	8	9	11	12	13	14	14	13	13	12	11	10	10	9
	0.4	0.4	2	4	5	7	9	11	12	14	15	16	16	16	16	15	15	14	13	12	11	10
	0.3	0.3	3	5	7	10	12	14	17	18	20	20	20	20	19	18	17	15	14	13	12	10
	0.2	0.2	3	6	9	12	15	18	21	24	25	25	25	23	22	20	19	17	15	14	13	11
	0.2	0.2	4	7	11	14	18	22	27	30	32	31	30	27	25	23	21	19	17	15	13	12
	0.2	0.2	4	8	12	17	21	27	32	38	40	39	35	31	28	25	22	20	18	16	14	13
	0.1	0.1	4	9	14	19	24	30	38	50	50	50	41	35	30	27	24	21	19	17	15	13
	0.1	0.1	5	9	14	20	26	33	41	50	50	50	45	37	32	28	25	22	20	17	15	14
	0.1	0.1	5	10	15	21	27	34	41	50	50	50	44	38	33	29	26	23	20	18	16	14
	0.1	0.1	5	10	15	21	27	34	42	50	50	50	44	38	33	29	26	23	20	18	16	14
	0.1	0.1	5	10	15	20	26	33	41	50	50	50	43	37	33	29	25	23	20	18	16	14
	0.1	0.1	5	9	14	19	25	31	39	50	50	50	41	36	31	28	25	22	20	17	15	14
	0.1	0.1	4	9	13	18	23	28	34	39	41	40	36	33	29	26	23	21	19	17	15	13
	0	0	4	8	12	16	20	25	29	32	34	33	32	29	27	24	22	20	18	16	14	13
	0	0	4	7	11	14	18	21	25	27	28	28	27	26	24	22	20	19	17	15	14	12
	0	0	3	6	10	13	16	19	21	23	24	24	24	23	22	20	19	17	16	14	13	12
	0	0	3	6	9	11	14	16	18	20	21	21	21	20	19	18	17	16	14	13	12	11
	0	0	3	5	7	10	12	14	16	17	18	18	18	18	17	16	15	14	13	12	11	10
	0	0	2	4	6	8	10	12	13	14	15	16	16	16	15	15	14	13	12	11	10	9
	0	0	2	4	6	7	9	10	12	12	13	14	14	14	13	13	12	12	11	10	9	9

Finite Difference

- Suppose that this represents a conductive rectangle held at 50V.

$$D_x = \epsilon_0 E_x$$

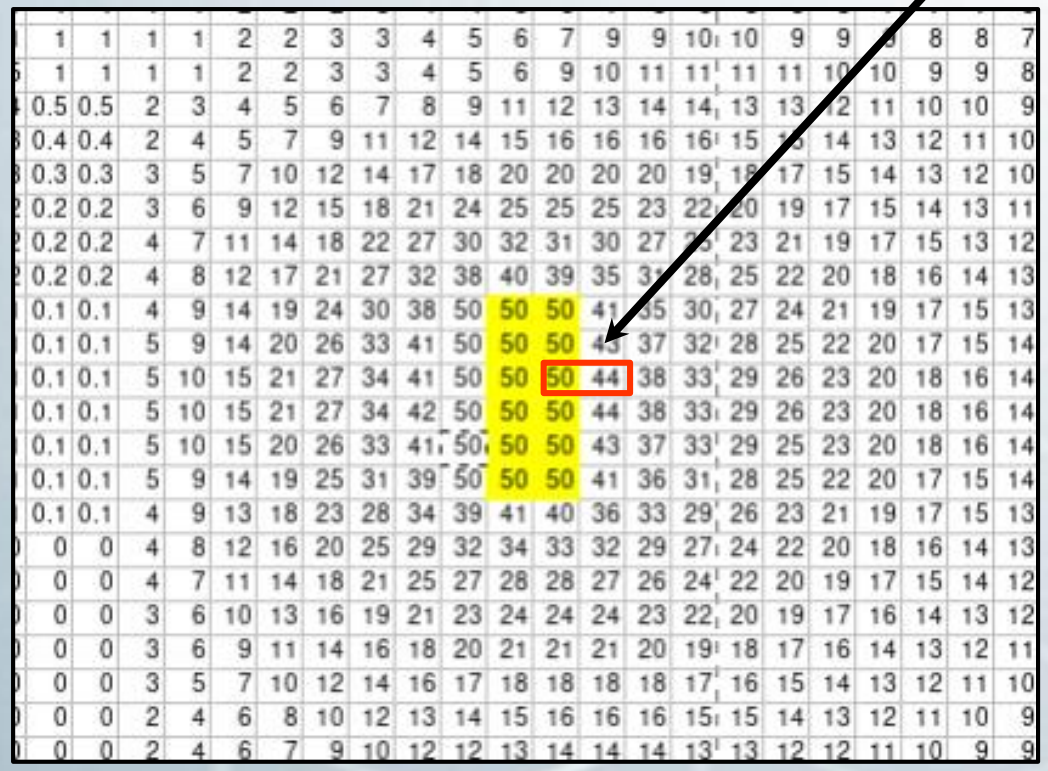


1	1	1	1	2	2	3	3	4	5	6	7	9	9	10	10	9	9	8	8	7	
1	1	1	1	2	2	3	3	4	5	6	9	10	11	11	11	11	10	10	9	9	8
0.5	0.5	2	3	4	5	6	7	8	9	11	12	13	14	14	13	13	12	11	10	10	9
0.4	0.4	2	4	5	7	9	11	12	14	15	16	16	16	16	15	15	14	13	12	11	10
0.3	0.3	3	5	7	10	12	14	17	18	20	20	20	20	19	18	17	15	14	13	12	10
0.2	0.2	3	6	9	12	15	18	21	24	25	25	25	23	22	20	19	17	15	14	13	11
0.2	0.2	4	7	11	14	18	22	27	30	32	31	30	27	25	23	21	19	17	15	13	12
0.2	0.2	4	8	12	17	21	27	32	38	40	39	35	31	28	25	22	20	18	16	14	13
0.1	0.1	4	9	14	19	24	30	38	50	50	50	41	35	30	27	24	21	19	17	15	13
0.1	0.1	5	9	14	20	26	33	41	50	50	50	45	37	32	28	25	22	20	17	15	14
0.1	0.1	5	10	15	21	27	34	41	50	50	50	44	38	33	29	26	23	20	18	16	14
0.1	0.1	5	10	15	21	27	34	42	50	50	50	44	38	33	29	26	23	20	18	16	14
0.1	0.1	5	10	15	20	26	33	41	50	50	50	43	37	33	29	25	23	20	18	16	14
0.1	0.1	5	9	14	19	25	31	39	50	50	50	41	36	31	28	25	22	20	17	15	14
0.1	0.1	4	9	13	18	23	28	34	39	41	40	36	33	29	26	23	21	19	17	15	13
0	0	4	8	12	16	20	25	29	32	34	33	32	29	27	24	22	20	18	16	14	13
0	0	4	7	11	14	18	21	25	27	28	28	27	26	24	22	20	19	17	15	14	12
0	0	3	6	10	13	16	19	21	23	24	24	24	23	22	20	19	17	16	14	13	12
0	0	3	6	9	11	14	16	18	20	21	21	21	20	19	18	17	16	14	13	12	11
0	0	3	5	7	10	12	14	16	17	18	18	18	18	17	16	15	14	13	12	11	10
0	0	2	4	6	8	10	12	13	14	15	16	16	16	15	15	14	13	12	11	10	9
0	0	2	4	6	7	9	10	12	12	13	14	14	14	13	13	12	12	11	10	9	9

Finite Difference

- Suppose that this represents a conductive rectangle held at 50V.

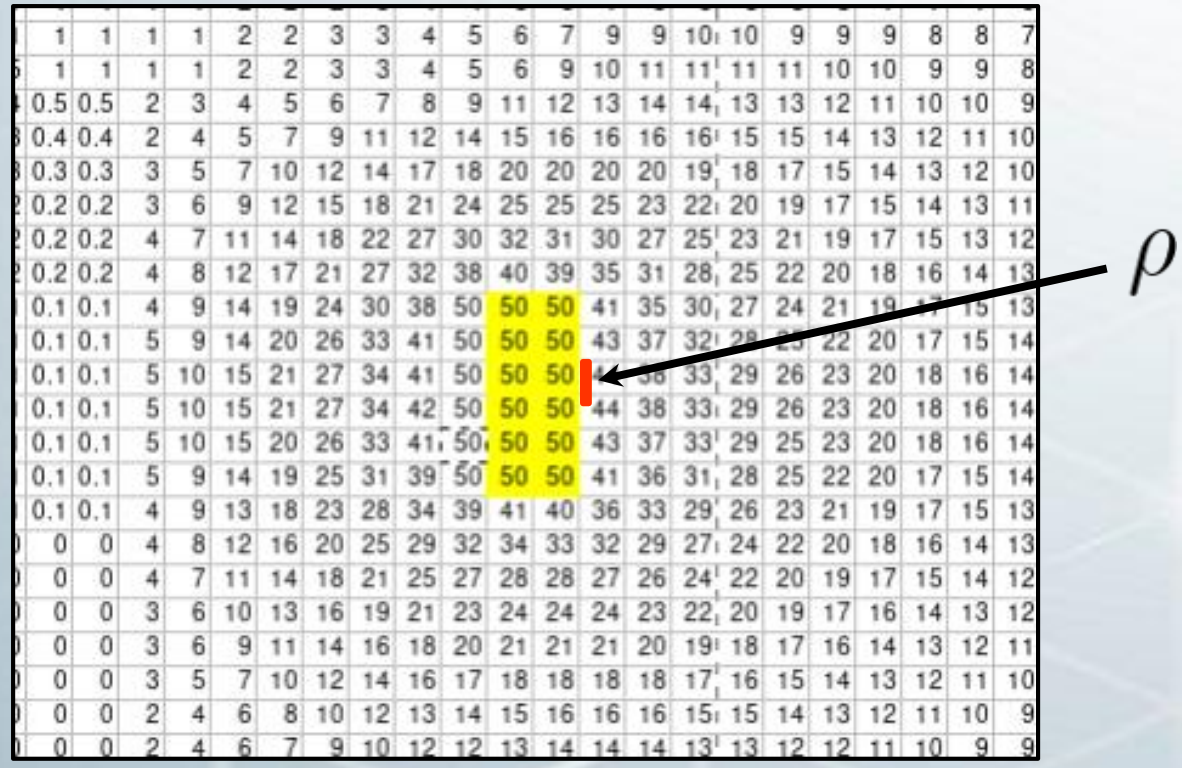
$$D_x = \rho$$



	1	1	1	1	2	2	3	3	4	5	6	7	9	9	10	10	9	9		8	8	7
	1	1	1	1	2	2	3	3	4	5	6	9	10	11	11	11	11	10	10	9	9	8
	0.5	0.5	2	3	4	5	6	7	8	9	11	12	13	14	14	13	13	12	11	10	10	9
	0.4	0.4	2	4	5	7	9	11	12	14	15	16	16	16	16	15	15	14	13	12	11	10
	0.3	0.3	3	5	7	10	12	14	17	18	20	20	20	20	19	18	17	15	14	13	12	10
	0.2	0.2	3	6	9	12	15	18	21	24	25	25	25	23	22	20	19	17	15	14	13	11
	0.2	0.2	4	7	11	14	18	22	27	30	32	31	30	27	25	23	21	19	17	15	13	12
	0.2	0.2	4	8	12	17	21	27	32	38	40	39	35	31	28	25	22	20	18	16	14	13
	0.1	0.1	4	9	14	19	24	30	38	50	50	50	41	35	30	27	24	21	19	17	15	13
	0.1	0.1	5	9	14	20	26	33	41	50	50	50	45	37	32	28	25	22	20	17	15	14
	0.1	0.1	5	10	15	21	27	34	41	50	50	50	44	38	33	29	26	23	20	18	16	14
	0.1	0.1	5	10	15	21	27	34	42	50	50	50	44	38	33	29	26	23	20	18	16	14
	0.1	0.1	5	10	15	20	26	33	41	50	50	50	43	37	33	29	25	23	20	18	16	14
	0.1	0.1	5	9	14	19	25	31	39	50	50	50	41	36	31	28	25	22	20	17	15	14
	0.1	0.1	4	9	13	18	23	28	34	39	41	40	36	33	29	26	23	21	19	17	15	13
	0	0	4	8	12	16	20	25	29	32	34	33	32	29	27	24	22	20	18	16	14	13
	0	0	4	7	11	14	18	21	25	27	28	28	27	26	24	22	20	19	17	15	14	12
	0	0	3	6	10	13	16	19	21	23	24	24	24	23	22	20	19	17	16	14	13	12
	0	0	3	6	9	11	14	16	18	20	21	21	21	20	19	18	17	16	14	13	12	11
	0	0	3	5	7	10	12	14	16	17	18	18	18	18	17	16	15	14	13	12	11	10
	0	0	2	4	6	8	10	12	13	14	15	16	16	16	15	15	14	13	12	11	10	9
	0	0	2	4	6	7	9	10	12	12	13	14	14	14	13	13	12	12	11	10	9	9

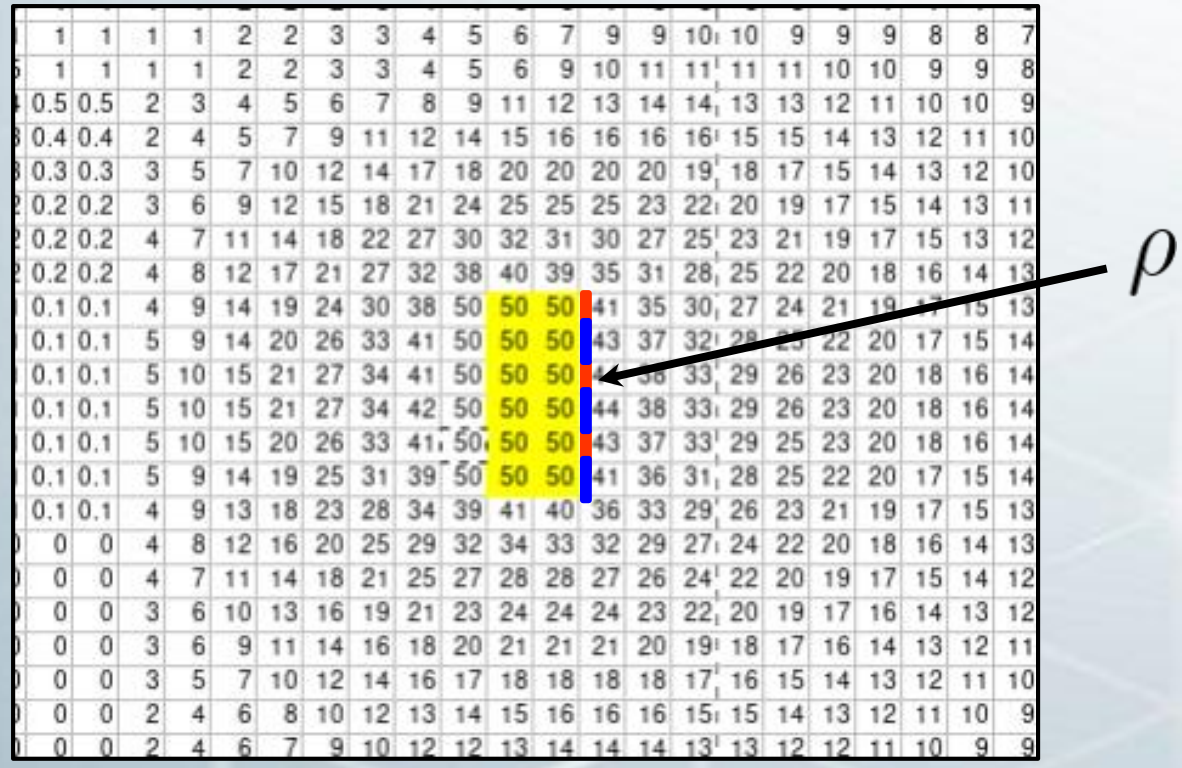
Finite Difference

- You now know the charge density on this surface of this cell of the conductor. You can do the same for adjacent cells.



Finite Difference

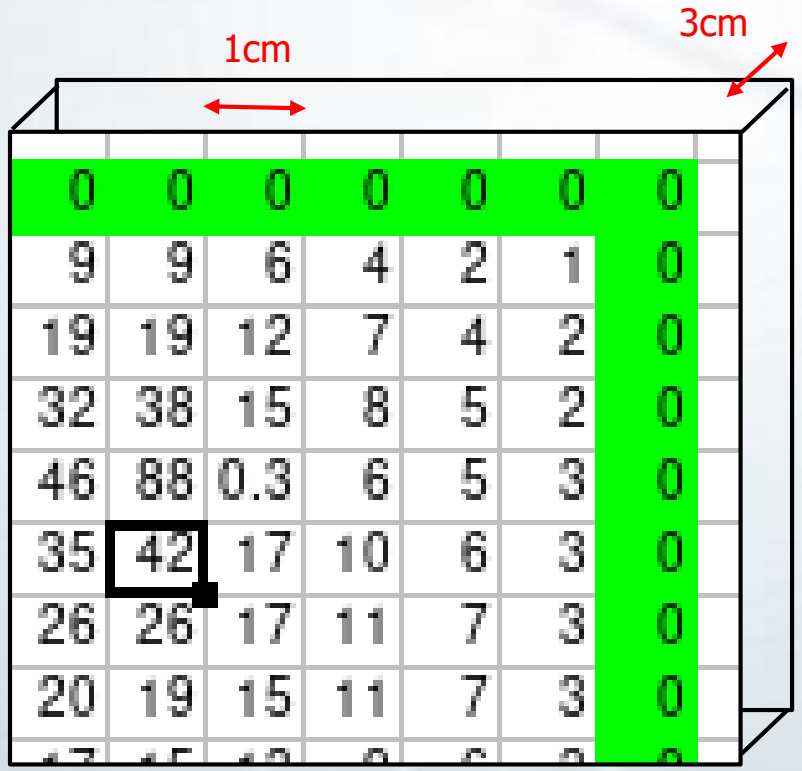
- You can then multiply the charge density on each of the cell surfaces below by its area to get the total charge on the right side of the conductor.



- How do you determine the area of this side of the conductor?

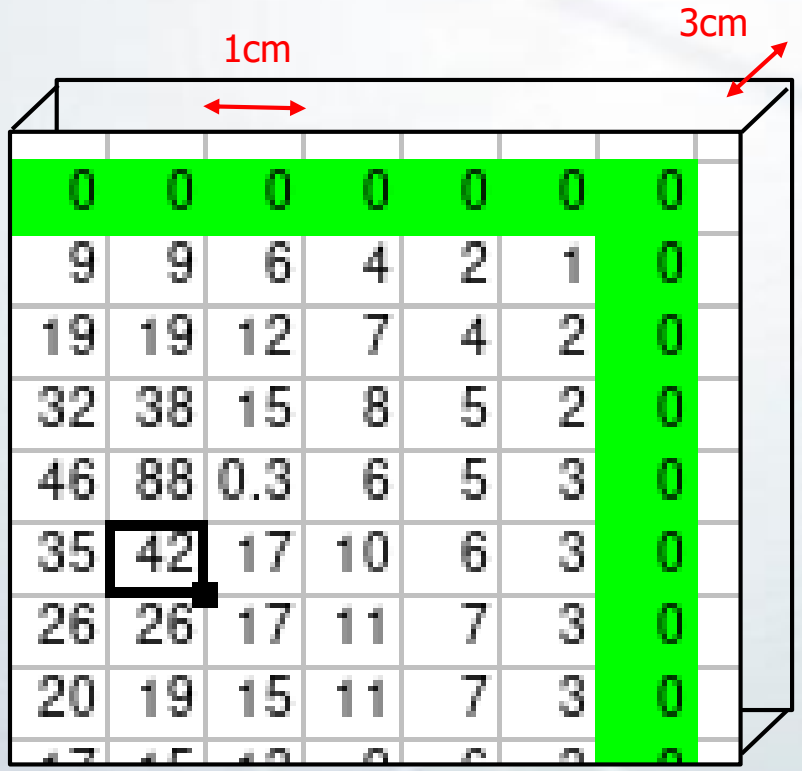
Finite Difference

- To calculate real quantities other than voltage, you must assign some dimension to each cell



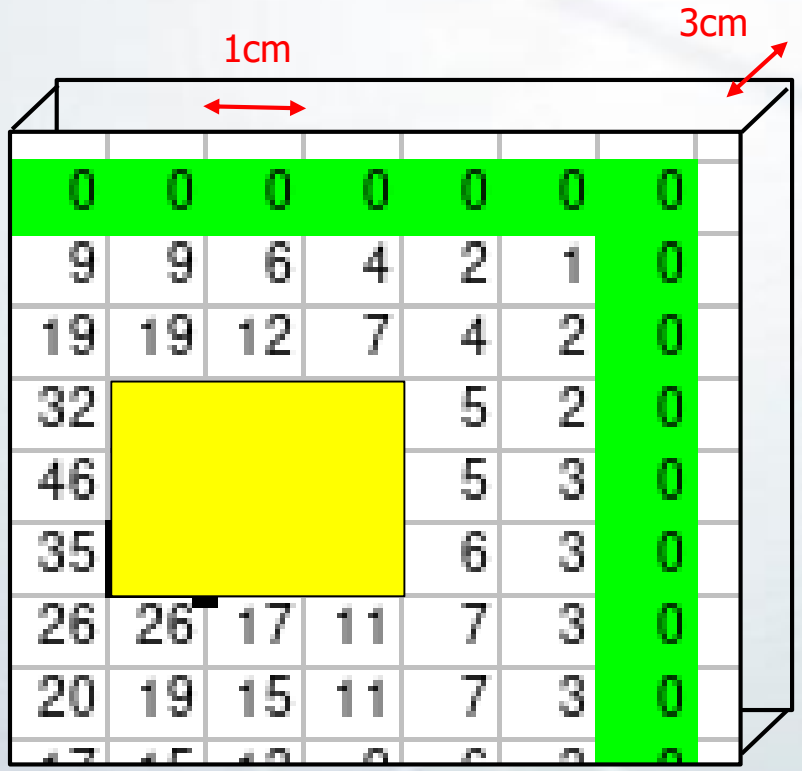
Finite Difference

- To represent true three-dimensional problem, each cell must also represent some depth (or represent per-unit length or depth).

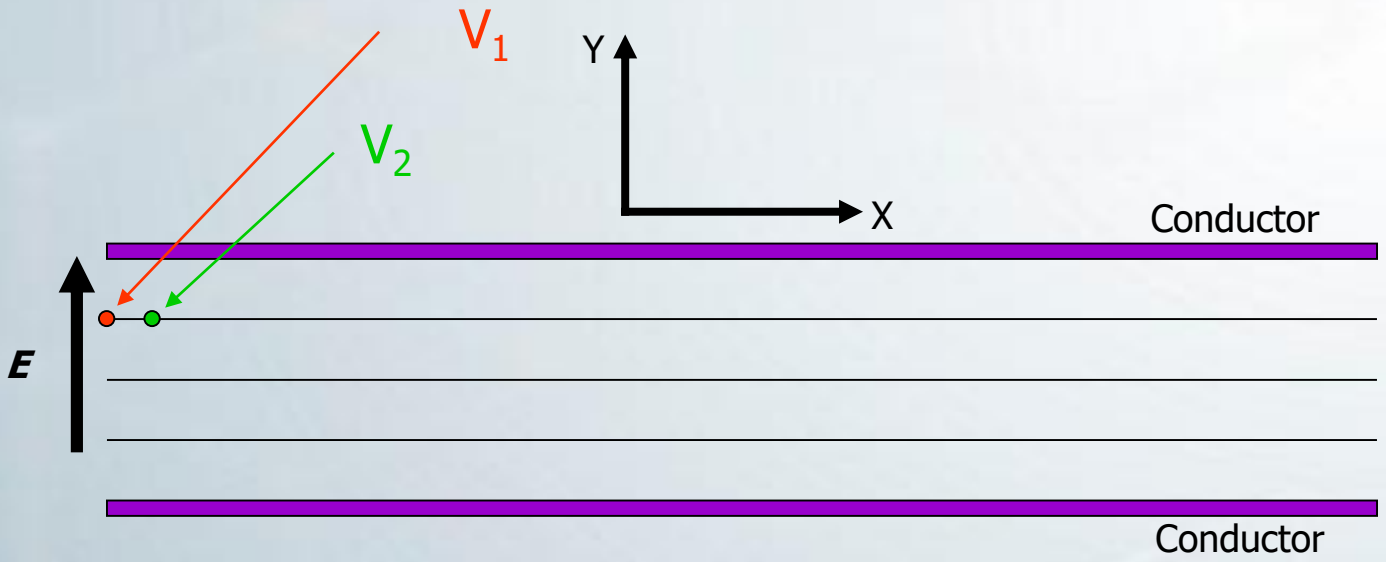


Finite Difference

- What is the area of the surfaces of this region parallel to the spreadsheet plane?

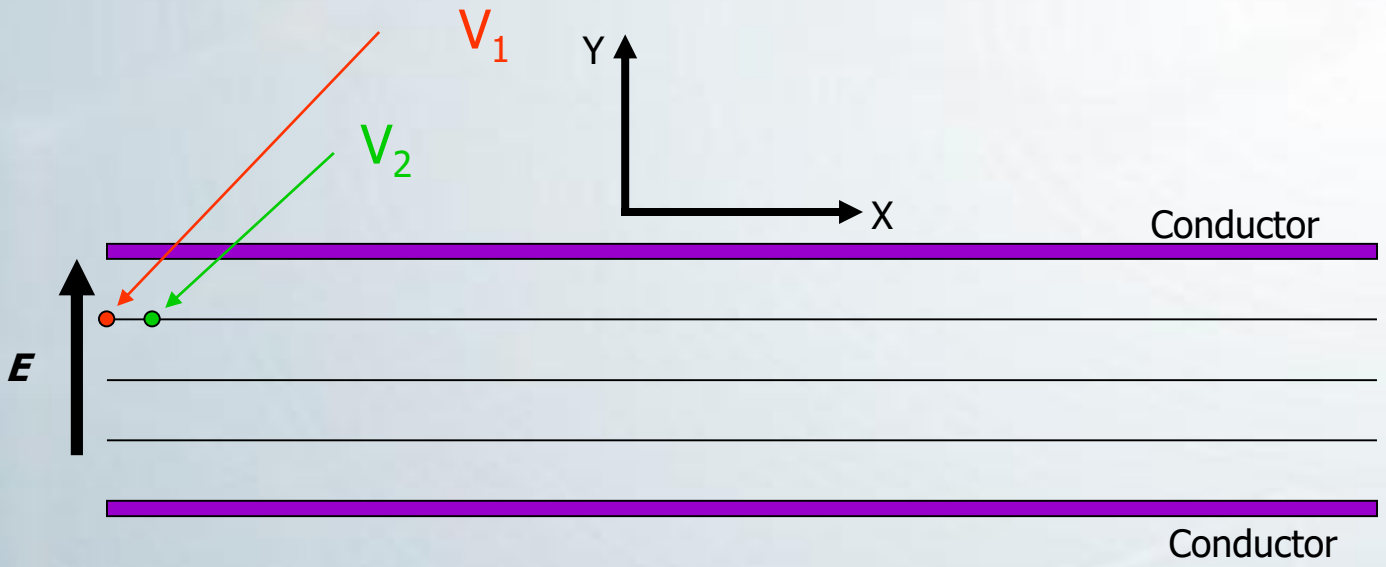


Finite Difference



- For a parallel plate capacitor, the side boundaries are open and the equipotential lines are horizontal.

Finite Difference



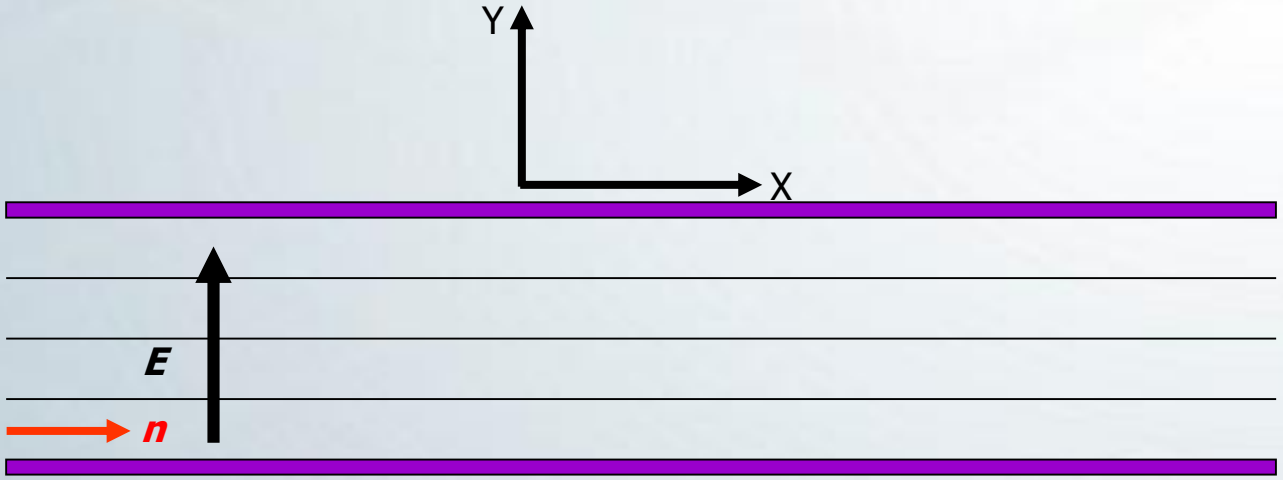
- For such voltage lines, the boundary voltage will equal the value of its immediate inside neighbor.

$$E_x = -\frac{\partial V}{\partial x} = 0$$

$$\therefore V_1 = V_2$$

Finite Difference

$$\frac{\partial V}{\partial n} = 0$$

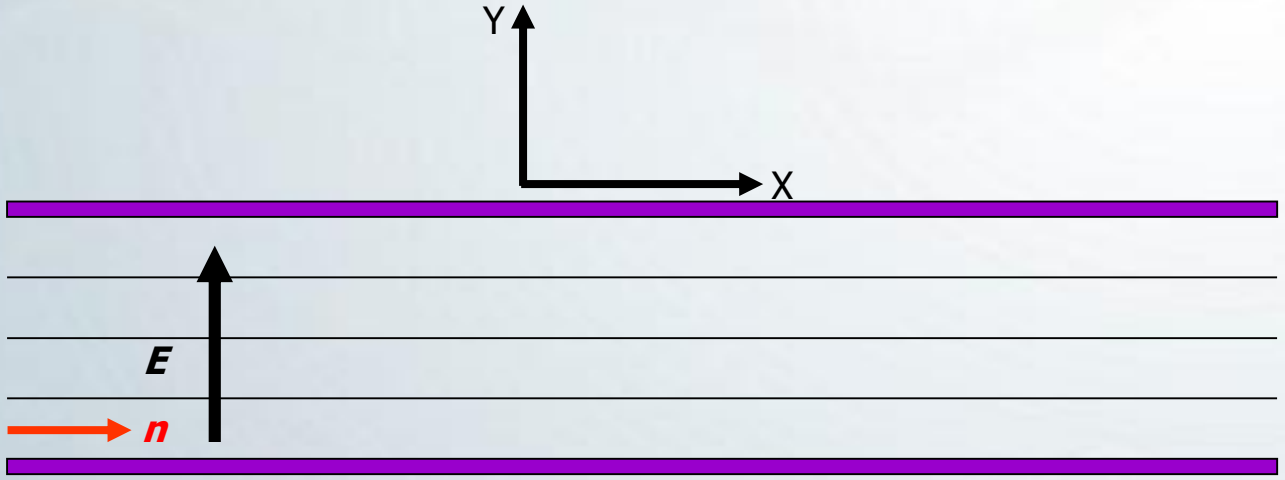


- Note that the condition that two points are on the same equipotential is the same as

$$D \bullet n = 0 \Leftrightarrow \hat{a}_x \bullet \varepsilon \nabla V = 0 \quad \text{written as} \quad \frac{\partial V}{\partial n} = 0$$

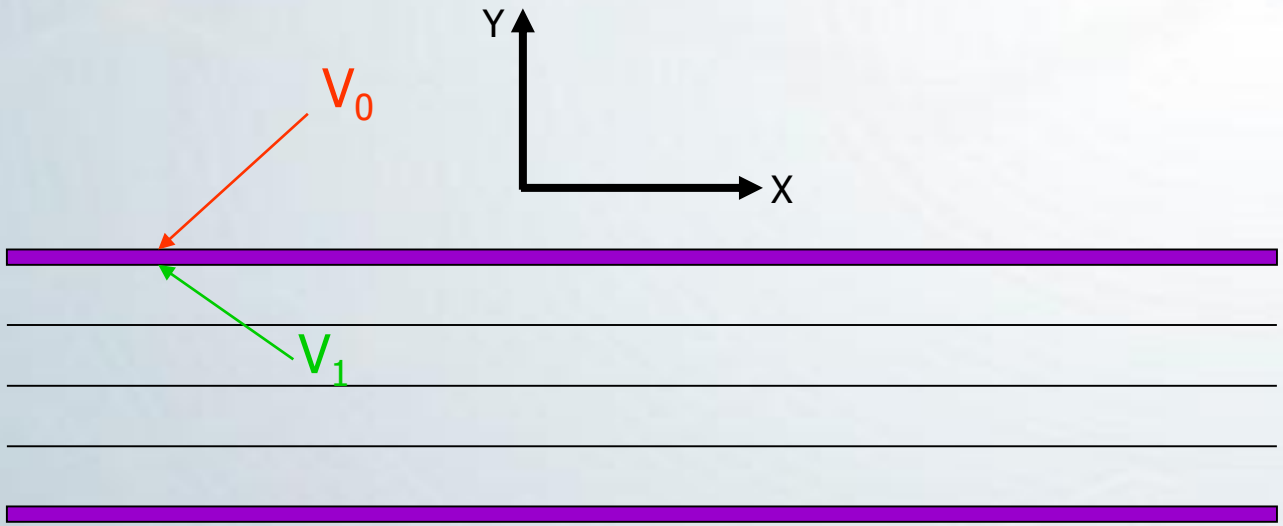
Finite Difference

$$\frac{\partial V}{\partial n} = 0$$



- This type of BC is called a Uniform Neumann Boundary Condition (from Mathematics). Specifies value of a derivative at the boundary.

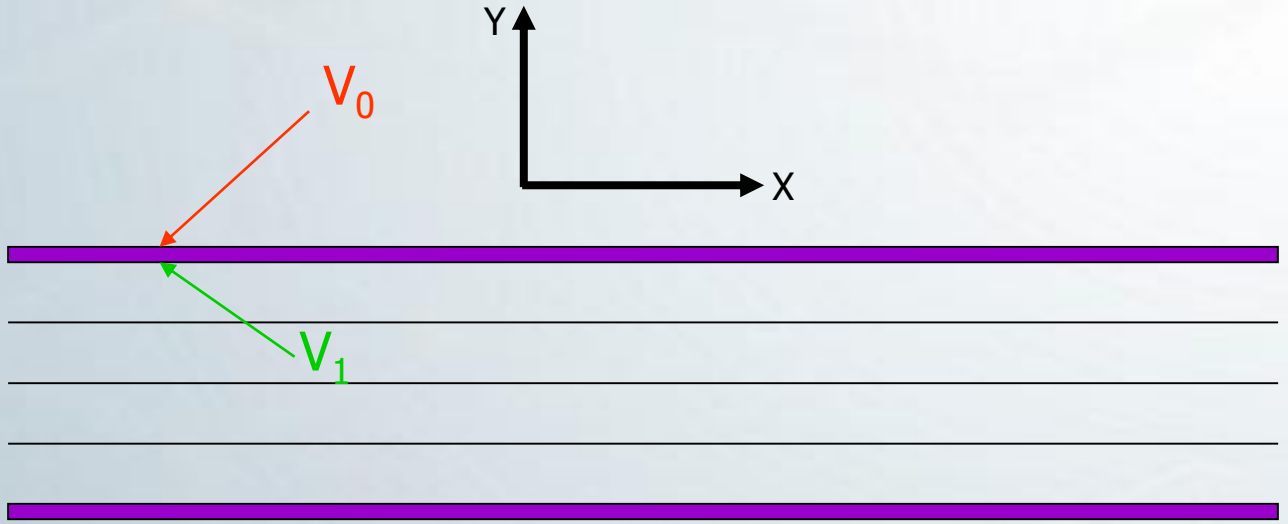
Finite Difference



- For completeness, we should also note that any part of the boundary that is given a **fixed voltage** has what is called a Dirichlet Boundary Condition

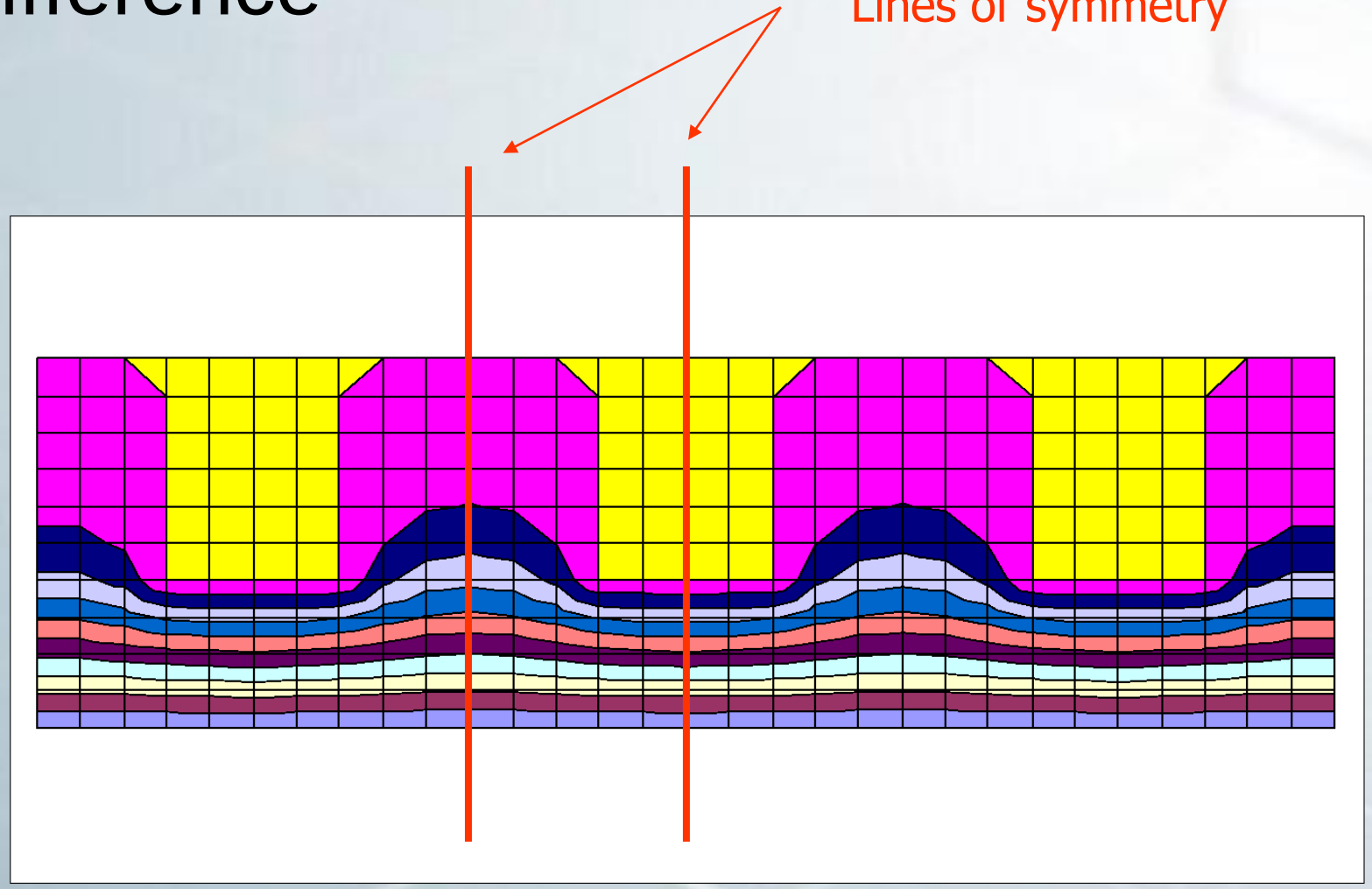
$$V_1 = V_0$$

Finite Difference



- At least one Dirichlet boundary condition needs to be specified for a unique solution

Finite Difference



- We can greatly reduce the work to find numerical solutions by using symmetry.

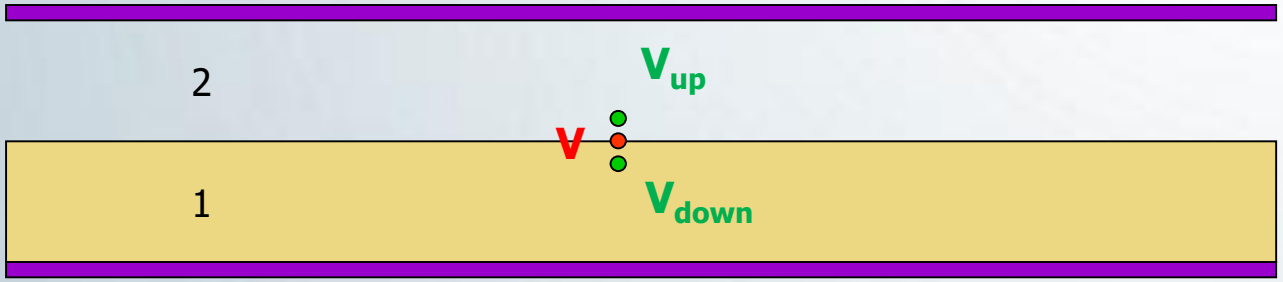
Finite Difference

- At a dielectric-dielectric boundary, the voltage is continuous, but the normal derivative is not. (Rate of change of voltage can be different.)

$$D_{n1} = D_{n2}$$

- What does this mean in terms of potential?

Finite Difference



- If there are two dielectrics, then the boundary condition at the interface must satisfy the dielectric-dielectric BC: $D_{n1} = D_{n2}$

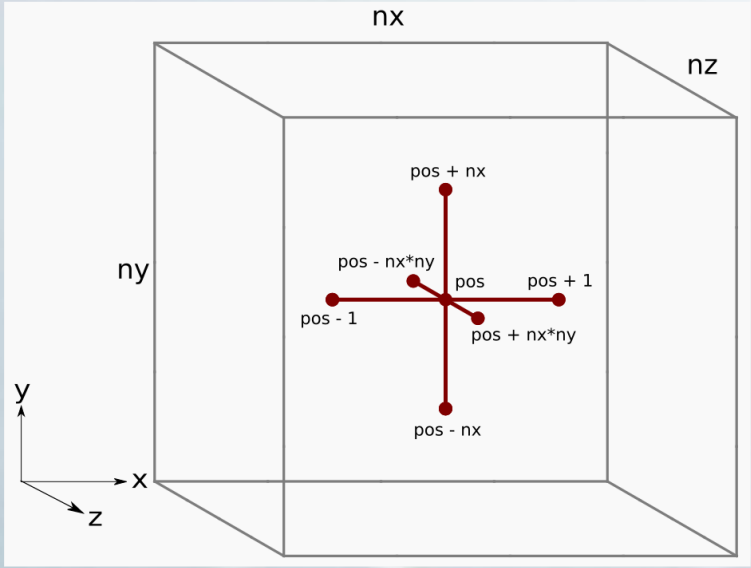
$$\varepsilon_1 \frac{V - V_{down}}{h} = \varepsilon_2 \frac{V_{up} - V}{h} \Rightarrow V = \frac{\varepsilon_1 V_{down} + \varepsilon_2 V_{up}}{\varepsilon_1 + \varepsilon_2}$$

Finite Element Method

Finite Element

- Finite Difference: generally, breaks a field region into a square grid

Finite Difference Method

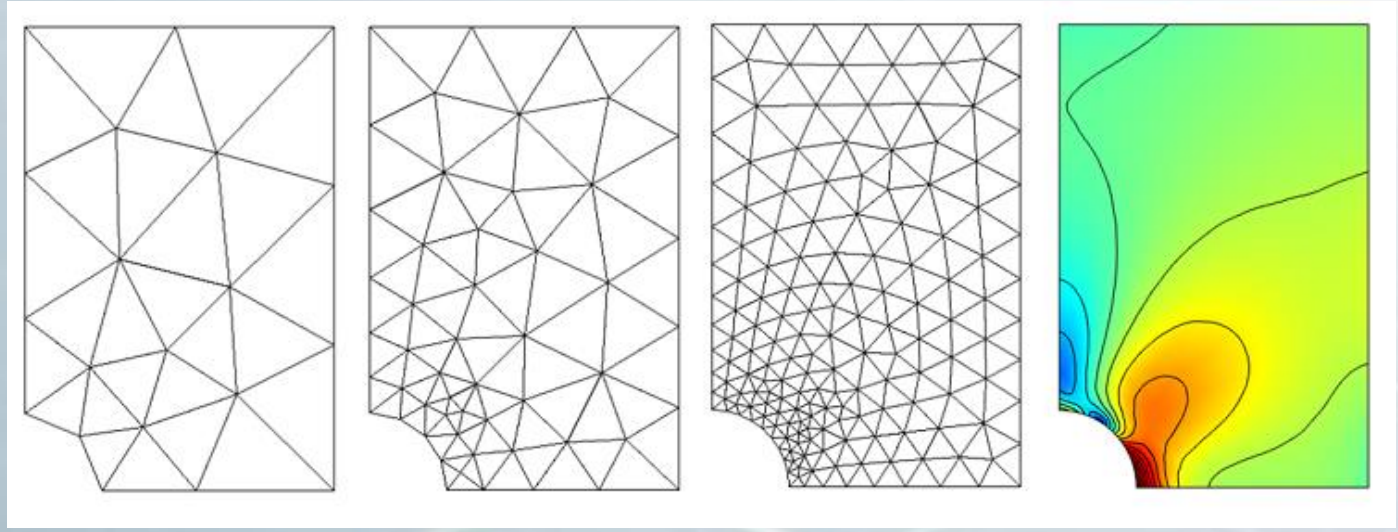


https://gpuopen.com/learn/amd-lab-notes/amd-lab-notes-finite-difference-docs-laplacian_part1/

Finite Element

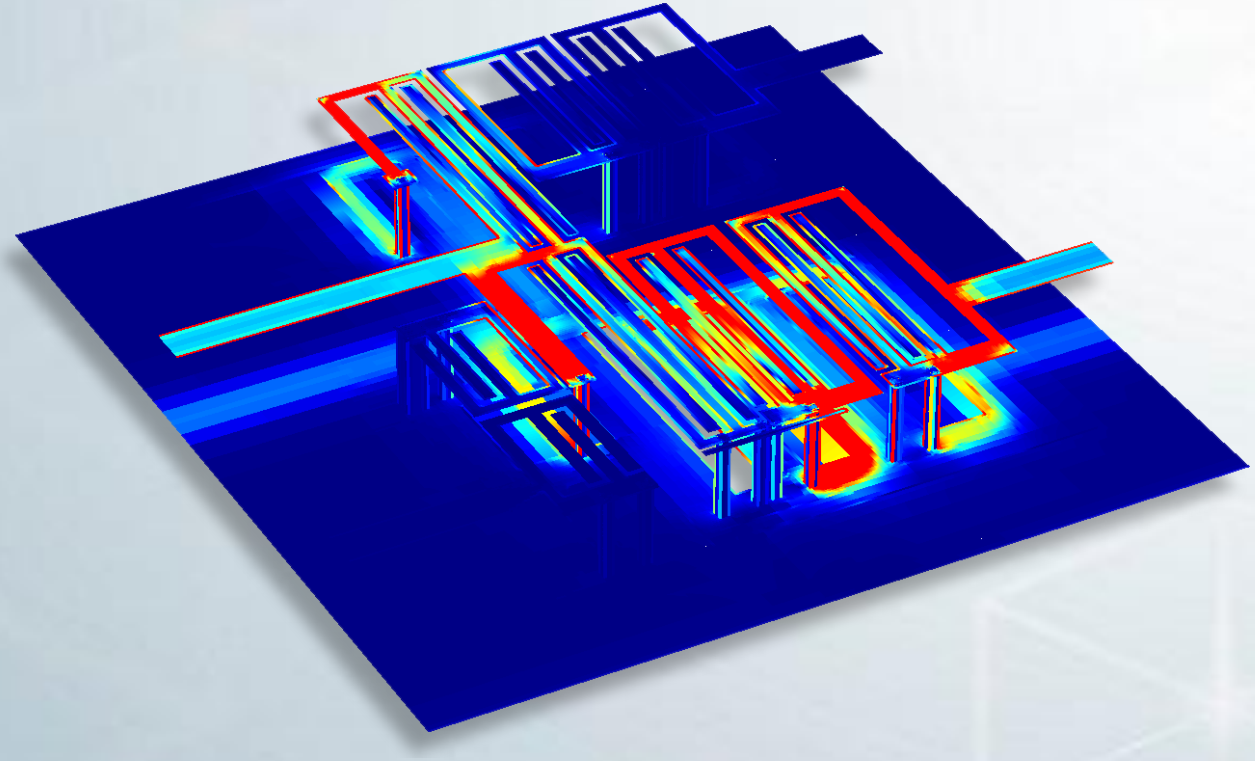
- Finite Element: can break a field region into subdivisions of any geometry

Finite Element Method



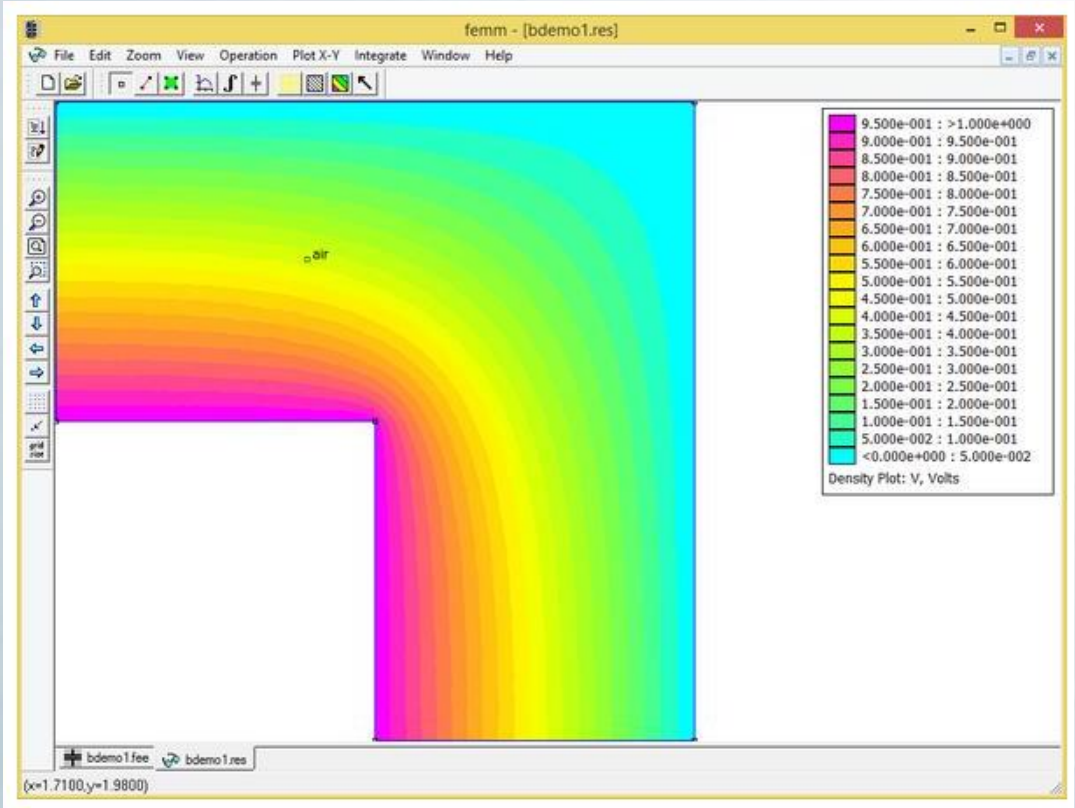
https://www.comsol.com/blogs/meshing-considerations-linear-static-problems

Finite Element



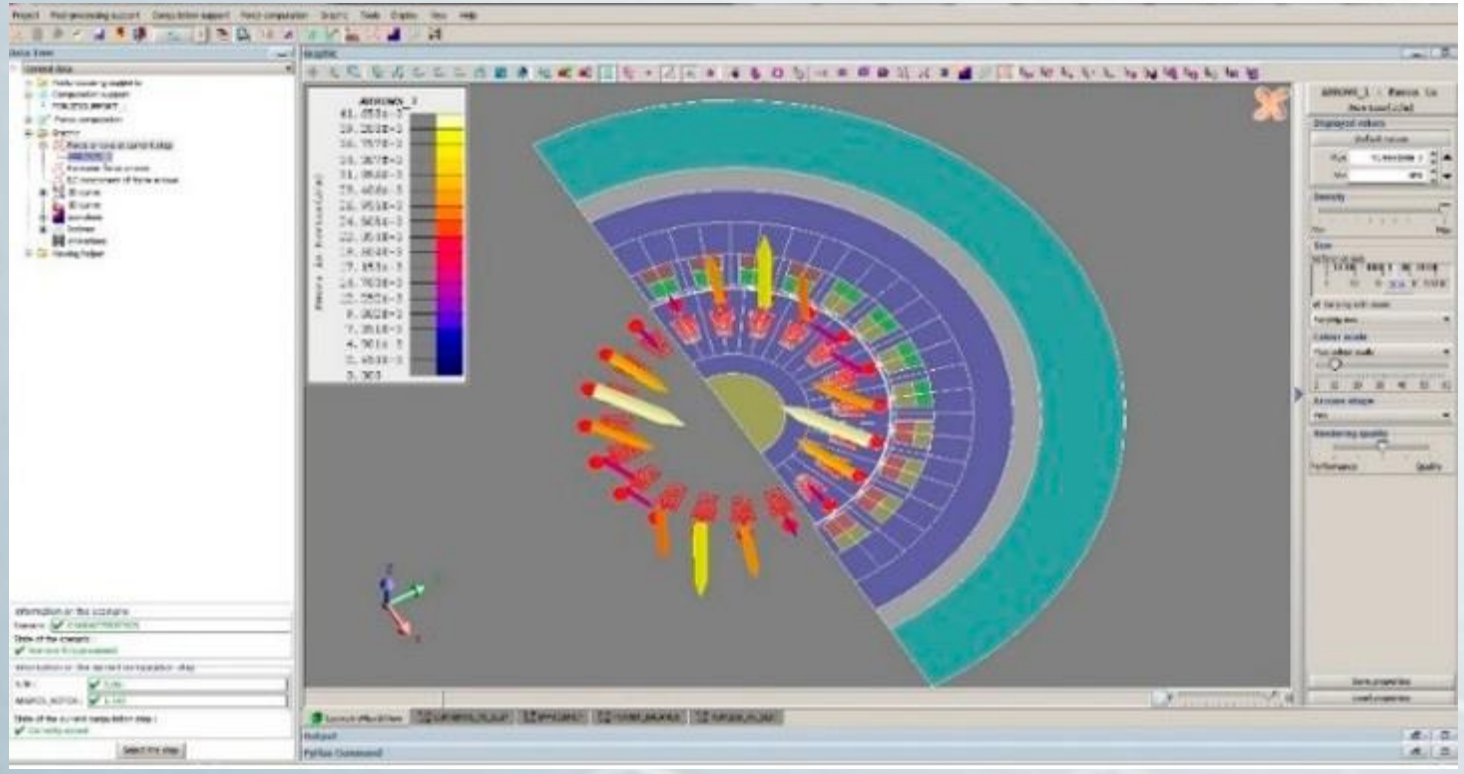
Sonnet (used for analog and digital circuit design applications)

Finite Element



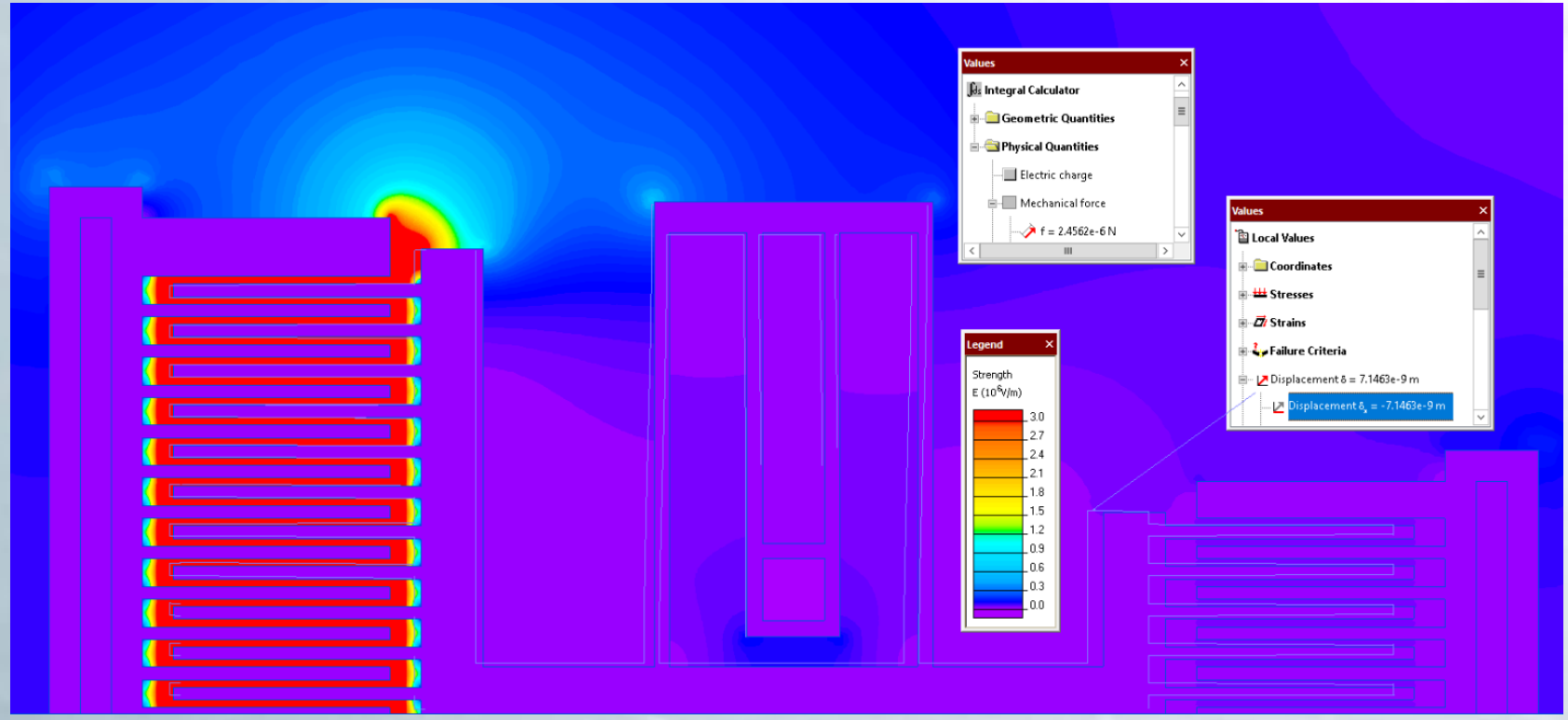
FEMM (open source, designed for magnetics but can be used for electrostatics)

Finite Element



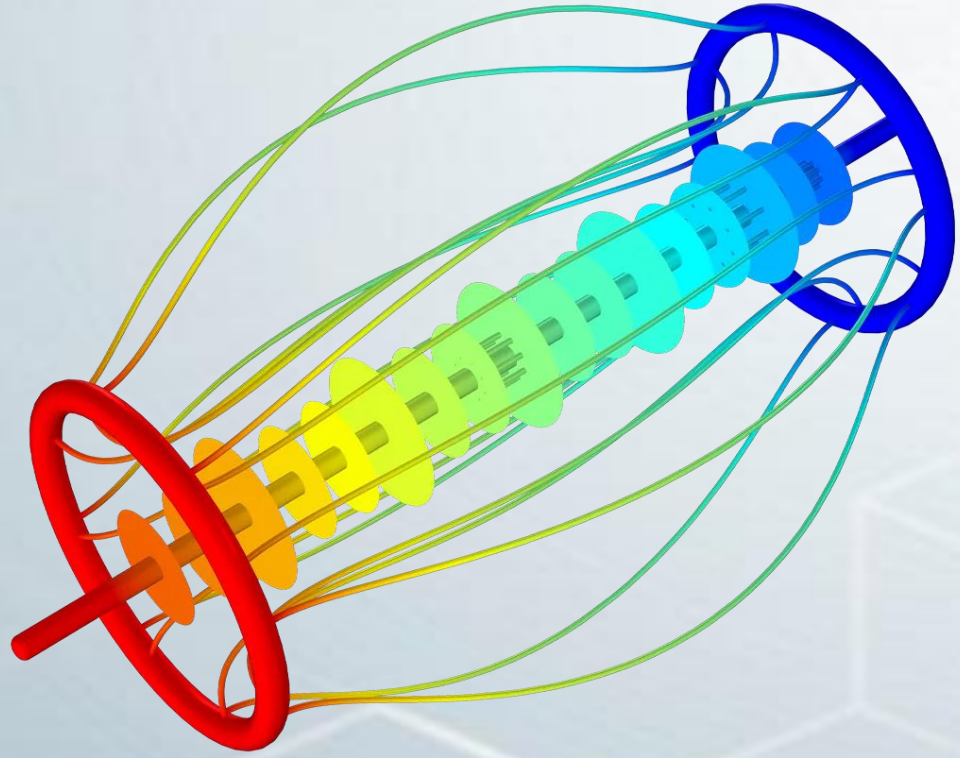
Altair Flux (wide range of electromagnetic capabilities)

Finite Element



QuickField

Finite Element



Comsol

Wrap Up

- HW6 due Wednesday
- Studio session next Wednesday is an extra set of office hours for exam preparation
- Exam 2 on Monday 10/26